

Apigee Edge Certificate Renewal Process Guide

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1. Introduction

This guide outlines the step-by-step process for renewing certificates in Apigee Edge, including the generation of a Certificate Signing Request (CSR) and a private key with an alias name, and the subsequent renewal or upload of the certificate into a TLS keystore.

2. Prerequisites

Before proceeding with the certificate renewal process, ensure you have the following:

- Access to the Apigee Edge management server.
- Appropriate administrative privileges in Apigee Edge to manage keystores and truststores.
- OpenSSL installed on your local machine or a server for generating CSRs and private keys.
- The existing keystore file (JKS or PKCS12 format) and its password.
- The alias name of the certificate to be renewed within the keystore.

- Network connectivity to the Apigee Edge management API.

3. Generating CSR and Key using OpenSSL

Generating a new CSR and private key is a crucial step in the certificate renewal process. The CSR contains information about your organization and domain, which will be used by the Certificate Authority (CA) to issue a new certificate. It is essential to associate the new key pair with an alias name within your keystore for proper management and identification.

3.1. Understanding Certificate Generation

3.2. Generating a New Key Pair and CSR using OpenSSL

To generate a new key pair and a CSR using OpenSSL, follow these steps:

1. **Open a terminal or command prompt.**
2. **Navigate to a directory where you want to store your key and CSR files.**
3. **Generate a new private key and CSR using the `openssl` command.** Replace the placeholders with your specific information:

```
bash openssl req -new -newkey rsa:2048 -nodes -keyout exelixi.com.key  
-out exelixi.com.csr
```

You will be prompted to enter information that will be incorporated into your certificate request. This is called a Distinguished Name (DN). You can leave some fields blank by entering `.`.

Example interaction:

```
```      Generating      a      2048      bit      RSA      private      key  
.....+++++
.....+++++ writing new private key to 'exelixi.com.key'
```

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You are about to be asked to enter information that will be incorporated into your certificate request. What you are about to enter is what is called a Distinguished

Name or a DN. There are quite a few fields but you can leave some blank For some fields there will be a default value, If you enter ' ', the field will be left blank.

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Country Name (2 letter code) [AU]:US State or Province Name (full name) [Some-State]:California Locality Name (eg, city) []:Irvine Organization Name (eg, company) [Internet Widgits Pty Ltd]:Exelixi Organizational Unit Name (eg, section) []:IT Common Name (e.g. server FQDN or YOUR name) []:exelixi.com Email Address []:iteai\_platform@exelixi.com

Please enter the following 'extra' attributes to be sent with your certificate request A challenge password []:changeit An optional company name []:uschlpcg2101:/tmp/Vinod/24OCT\_2023> ls -lrt ```

This command will create two files: a private key ( `.key` ) and a Certificate Signing Request ( `.csr` ). You will submit the `.csr` file to your Certificate Authority (CA) to obtain a new signed certificate.

## 4. Renewing Certificate in TLS Keystore

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Once you receive the renewed certificate from your Certificate Authority (CA), you need to import it into your existing TLS keystore in Apigee Edge. This process involves uploading the new certificate and associating it with the correct alias.

### 4.1. Understanding Apigee Edge Keystores

In Apigee Edge, keystores are used to store TLS certificates and their corresponding private keys. These keystores are then associated with virtual hosts or target endpoints to secure API traffic. When renewing a certificate, you are essentially updating the certificate entry within an existing keystore.

### 4.2. Importing the Renewed Certificate

There are two primary methods to renew or upload a certificate in a TLS keystore in Apigee Edge: using the Apigee Edge UI or using the Apigee Management API.

#### 4.2.1. Using Apigee Edge UI

1. Log in to the Apigee Edge UI.

2. **Navigate to** `Admin > TLS Keystores` (or `Admin > Environments > TLS Keystores` depending on your Apigee Edge version).
3. **Select the keystore** where you want to renew the certificate.
4. **Click on the** `+ certificate` **button** or an equivalent option to add a new certificate.
5. **Choose the option to upload a certificate.** You will typically have options to upload a certificate file and its private key, or a certificate chain.
6. **Provide the following information:**
  - **Alias:** Enter the same alias name that you used when generating the CSR (e.g., `mycertalias`). This is crucial for renewing an existing certificate.
  - **Certificate File:** Upload the new certificate file provided by your CA (usually a `.cer`, `.crt`, or `.pem` file).
  - **Private Key File (if applicable):** If your CA provided the certificate and key separately, upload your private key file (the one generated with the CSR).
  - **Passphrase (if applicable):** If your private key is encrypted, provide the passphrase.
7. **Save the changes.** Apigee Edge will validate the certificate and import it into the keystore under the specified alias. If the alias already exists, it will effectively renew the certificate associated with that alias.

## 4.3. Verifying the Renewed Certificate

After importing the renewed certificate, it's crucial to verify that it has been successfully updated and is being used by your Apigee proxies. You can do this by:

- **Checking the certificate details in the Apigee Edge UI:** Navigate to the keystore and inspect the certificate associated with the alias. Verify the new expiry date and other details.
- **Testing your API endpoints:** Make API calls to the endpoints that use this keystore and ensure that the new certificate is being presented. You can use `curl -v` to inspect the certificate chain.

## 5. Conclusion

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Renewing certificates in Apigee Edge is a critical maintenance task to ensure the continued security and availability of your APIs. By following the steps outlined in this guide, you can effectively generate CSRs, manage key pairs, and seamlessly renew certificates in your TLS keystores, whether through the Apigee Edge UI or the Management API. Regular certificate management practices are essential for maintaining a robust and secure API ecosystem.

### 4.2.2. Restarting Message Processors and Routers

After successfully importing the renewed certificate into the TLS keystore, it is crucial to restart the Apigee Message Processors and Routers to ensure that they pick up the new certificate. This step is essential for the changes to take effect and for your APIs to start using the renewed certificate.

To restart the Message Processors and Routers, you typically need to access the Apigee Edge nodes and execute specific commands. The exact commands may vary depending on your Apigee Edge installation (on-premise or cloud-based) and version. Consult your Apigee Edge documentation for precise instructions.

For on-premise installations, you might use commands similar to these on each Message Processor and Router node:

```
/opt/apigee/apigee-service/bin/apigee-service edge-message-processor restart
/opt/apigee/apigee-service/bin/apigee-service edge-router restart
```

**Important:** Ensure that you restart all relevant Message Processors and Routers in your Apigee Edge deployment to avoid any inconsistencies or service disruptions.